

EMPIRICAL ARTICLE

Associations between social media and crystallized and fluid performance trajectories in early adolescence

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Abstract

Longitudinal evidence on social media use and cognitive abilities trajectory coupling across early adolescence is sparse. In this study, data from the ABCD Study (Baseline $N \sim 12,000$; $M_{age} = 9.9$; White = 52.0%, Hispanic = 20.3%, Black = 15.0%, Asian = 2.1%, Other = 10.5%; Female = 47.8%) were used to examine whether growth processes in social media use and cognitive performances (i.e., crystallized abilities, inhibitory control/attention, and processing speed) were associated over 4 years in early adolescence. Parallel process latent growth curve models revealed that the trajectory of social media use was negatively coupled with crystallized and inhibitory control/attention performance trajectories, indicating potential trade-offs. Conversely, the trajectories of social media and processing speed were positively coupled, possibly due to practice-based transfer.

KEYWORDS

adolescence, cognitive functioning, executive functioning, social media use

INTRODUCTION

The emergence of social media use in early adolescence is both historically novel yet contemporarily normative, prompting considerable research interest aimed at understanding the developmental implications of this context. While a substantial portion of social media research has focused on its effect on mental health, its role in cognitive development is less often studied. Online participation in social media typically begins in early adolescence and increases markedly in the following years (Rideout et al., 2022). Concurrently, early adolescence is also a period of significant cognitive development (Larsen & Luna, 2018; Steinberg, 2005), rendering the adolescent brain especially vulnerable to environmental influences on cognitive formation (Ferschmann et al., 2022; Murty et al., 2016; Spear, 2013). Thus, it is critical to elucidate the nature of prospective associations between time spent using social media and cognitive development. Early cross-sectional evidence suggests that spending more hours on social media is negatively related to general cognitive development (Walsh et al., 2020). Yet, cognitive functioning comprises separable domains, including fluid and crystallized abilities (Cattell, 1987), and the longitudinal

associations between social media use and these abilities across early adolescence remain poorly understood. In the reverse direction, it may likewise be that differences in cognitive abilities influence trajectories of social media behavior. However, to date, bidirectional tests of these longitudinal associations have yet to be conducted. To address these gaps, the current study examines how time spent using social media and performance on fluid and crystallized growth are linked across 4 years in early adolescence.

Social media in early adolescence

Social media has become a primary leisure activity for adolescents, and a substantial portion of their social life now takes place online (Kennedy & Lynch, 2016; Tsitsika et al., 2014). Social media comprises a range of online activities including direct messaging, posting photos and videos, and viewing content posted by peers or individuals in broader online communities. Empirical studies suggest youth typically begin to use social media in early adolescence, with 18% of 8- to 12-year-olds using social media every day (Rideout et al., 2022). By middle adolescence, 84% of youth report being active on social media, logging in for

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an average of 1 h and 27 min per day based on self-reports (Rideout et al., 2022). The recent rise in social media use has generated substantial interest regarding its impact on youth development.

Cognitive development

Cognition broadly refers to a set of interrelated executive abilities such as working memory, reasoning, information processing, attention, language comprehension, and verbal abilities (Luciana et al., 2018). Youth are not born with full adult cognitive capacities and instead acquire them gradually throughout the first decades of life (Wood et al., 2004, p. 70). The onset of adolescence heralds a particularly extensive period of maturation of brain regions known to underlie growth in cognitive abilities (Paus, 2005). Neurobiologically, this process is mediated through increased rates of synaptic pruning and myelination which increase communication efficiency between neural structures (Blakemore & Choudhury, 2006), particularly in frontal lobe regions and associated areas (Murty et al., 2016). These neural changes make early adolescence a period of heightened vulnerability to impacts on cognitive developmental trajectories (Luciana, 2013) making it critical to identify contexts that confer risk during this period.

Although approaches to measuring cognitive development vary, hierarchical approaches rooted in two-component models of intelligence have shown considerable durability (Blair, 2006; Horn & Cattell, 1966; Luciana et al., 2018; Schneider & McGrew, 2012). This premise assumes that general cognitive ability is underlain by two (Horn & Cattell, 1966) broad abilities that include fluid and crystallized cognition. Fluid abilities refer to the cognitive capacity to integrate new information and solve novel problems. Such abilities include sustaining attention, inhibitory control, task switching, memory recall, and information processing (Akshoomoff et al., 2013). In contrast, crystallized abilities refer to skills acquired from one's environment, such as factual knowledge, verbal abilities, cultural competencies, and reading comprehension (Akshoomoff et al., 2013). This approach is reflected in common instruments for measuring cognitive ability, such as the NIH Toolbox, which features fluid and crystallized ability composite scores (Slotkin et al., 2012).

The transition to adolescence marks the incipience of significant neural maturation that underlies a host of cognitive changes (Steinberg, 2005). This is particularly evident in prefrontal brain regions where increased synaptic pruning and myelination contribute to an acceleration in cognitive ability development that continues into early adulthood (Blakemore & Choudhury, 2006). While growth in both fluid and crystallized abilities has been observed empirically, the latter has been shown to outpace the former throughout adolescence (Schroeders et al., 2015).

Although cognitive abilities are highly heritable in general across the lifespan (Plomin et al., 2013), they are also

subject to environmental influences (Nisbett et al., 2012). Fluid abilities have traditionally been considered to be more heritable than crystallized abilities (Hoekstra et al., 2007; van Soelen et al., 2011), yet evidence is mixed on this point (Kan et al., 2013; Loughnan et al., 2023; Procopio et al., 2022). For crystallized abilities, despite substantial genetic contributions, these skills are widely viewed to develop over time through cumulative environmental exposure (Freeman, 1983; Horn, 1968) shaped by “formal and informal educational and general life experiences” (McGrew, 2009; Tucker-Drob & Briley, 2014). Further evidence from early adolescent samples indicated that environmental factors account for a substantial portion of variability in verbal IQ, a crystallized ability, but not nonverbal IQ (Hoekstra et al., 2007; van Soelen et al., 2011). For fluid competencies, substantial evidence for training effects suggests this type of ability can be influenced by one's environment (see Nisbett et al., 2012, p. 137 for review).

The impact of social media on cognitive performance

Empirical evidence suggests that cognitive performances are associated with time spent using screens (Liu et al., 2022; Madigan et al., 2019; Neophytou et al., 2021). Moreover, specific types of screen use have been shown to exhibit distinct associations with cognitive abilities. For example, while playing video games has been found to have cognitive benefits for general cognitive abilities (Sauce et al., 2022; Walsh et al., 2020), social media use was either unrelated (Sauce et al., 2022) or negatively associated (Walsh et al., 2020) with such outcomes. However, longitudinal studies on the specific impacts of social media use on fluid and crystallized abilities in early adolescence are scarce.

Much of what is known regarding the impact of social media on fluid abilities derives from late adolescent and early adult studies. Negative, albeit very small, cross-sectional associations have been found between social media use and fluid abilities (Stieger & Wunderl, 2022) and media multitasking and working memory (Cain et al., 2016) in middle adolescence. Moreover, one literature review noted that there is ample evidence for *acute* impacts of social technology's presence (e.g., in the presence of a smartphone) on state-like attention shifting, working memory, and information processing in adolescents and adults (Wilmer et al., 2017). Further, two studies found negative cross-sectional associations between problematic social media use and response inhibition performance in young adults (Gou et al., 2024; Reed, 2023). However, one adult study found no significant association between social media and short-term memory performance (Almarzouki et al., 2022) and further evidence that heavy social media use leads to durable changes in fluid abilities has been found to be weak or nonexistent in early adulthood (Lara & Bokoch, 2021; Reed, 2023). In summary, while acute effects

on state-like fluid performance have been established (Wilmer et al., 2017), convincing directional evidence that social media use leads to lasting impacts on fluid competency in early adolescence is lacking.

Evidence for the effects of social media on early adolescents' crystallized abilities is similarly scarce. Cross-sectional studies suggest that social media is related to worse academic performance (Junco, 2012b; Kirschner & Karpinski, 2010), and text messaging is negatively associated with grammar skills in early adolescence (Cingel & Sundar, 2012). Furthermore, media multitasking involving social media was related to worse middle adolescent English test performance, but not vocabulary scores (Cain et al., 2016). In contrast, Facebook use was cross-sectionally related to better middle adolescent verbal ability in another small study (Alloway et al., 2013). However, it is noteworthy that social media behavior has changed (e.g., TikTok and Instagram were launched in 2016 and 2020, respectively) and the rates of social media use have increased in adolescence since these studies were conducted (Rideout et al., 2022). High-quality longitudinal studies in early adolescent youth on this topic have yet to emerge.

Several features of social media might impact the development of crystallized abilities. Social media platforms frequently use algorithms to curate content based on users' preferences (Kim, 2017), potentially displacing exposure to alternate activities, concepts, and vocabulary. This is especially plausible given evidence that social media use displaces time spent reading (Twenge et al., 2019). Additionally, instant messaging notifications, a common feature of social media, can interrupt attention and hinder one's ability to consolidate knowledge from their surroundings (Horvitz & Cutrell, 2001). Indeed, frequent notifications and the constant presence of mobile devices have been linked to a reduced capacity for environmental learning (Wilmer et al., 2017). Over time, these processes may undermine knowledge accumulation and impede the development of these crystallized abilities.

The impact of cognitive abilities on time spent using social media

Although the cognitive impacts of social media have garnered some attention, the reverse effect has been less regarded. However, it is plausible that cognitive ability might influence social media habits. Indeed, some supporting evidence exists. For example, attention problems, which are related to several fluid abilities (Brydges et al., 2021), have been linked to an increased risk of social media addiction (Brydges et al., 2017; Thorell et al., 2022). This suggests that there may be some social media selection effects that vary by fluid abilities. Yet, empirical evidence and theory on whether crystallized abilities influence social media levels are lacking. Testing whether these processes covary over time will provide important developmental insights on this topic.

The present study

Longitudinal tests of bidirectional associations between time spent using social media and cognitive skills are lacking. The current study aims to address these gaps by testing associations between social media hours and cognitive performance (i.e., crystallized and fluid) trajectories across early adolescence (i.e., three time points; Year 0 $M_{age} = 9.9$, Year 2 $M_{age} = 12.0$, Year 4 $M_{age} = 14.1$). To this end, we employ parallel process latent growth curve models to determine whether the linear trajectories (i.e., slopes) of these constructs are linked, while controlling for youth sex and parental education. Further, to account for variance associated with other screen activities, we also control for time spent using video games, online streaming, and television.

With respect to crystallized abilities, we hypothesize there will be a significant negative slope-slope association such that greater increases in social media hours will be linked to less growth in crystallized abilities (H1), consistent with evidence showing that social media engagement may crowd out structured academic activities (Twenge et al., 2019). Turning to fluid abilities, although prior experimental studies have shown that the presence of social media may acutely impact fluid abilities (Wilmer et al., 2017), findings from observational studies have not consistently demonstrated a developmental link (Gou et al., 2024; Lara & Bokoch, 2021; Reed, 2023). Accordingly, we hypothesize that there will be no association between social media and fluid abilities trajectories (H2). Collectively, these tests will help clarify whether adolescents' social media behaviors and cognitive trajectories evolve in tandem during this critical developmental period.

METHOD

We used data from the Adolescent Brain Cognitive Development (ABCD) Study (5.1 release) to address study aims. The ABCD study recruited nearly 12,000 adolescents aged 9–10 across 21 sites beginning in 2015 with follow-up data collection planned every 6 months for a total of 10 years (Casey et al., 2018). Overall, the data collected consist of neuroimaging, cognitive tasks, and survey data. Data collection sites were selected based on research expertise and neuroimaging equipment availability and were thus largely restricted to urban areas. Data for the current study was from the 5.1 release.

Participants

The current study used data from baseline (Y0: $N = 11,876$; $M_{age} = 9.9$), the 2-year follow-up (Y2: $N = 100,972$; $M_{age} = 12.0$), and the 4-year follow-up (Y4: $N = 4754$; $M_{age} = 14.1$). Per design, the ABCD Study releases roughly half of the most recent wave's data each year. The sample was designed to be

a representative sample of US adolescents (White=52.0%, Hispanic=20.3%, Black=15.0%, Asian=2.1%, Other=10.5%; Female=47.8%). Moreover, roughly 41% of children were from families with an annual income of \$100,000 or higher and 73% were from married, two-parent households at baseline (Heeringa & Berglund, 2020).

Measures

The current study analyzed survey and task-based cognitive performance data at three time points across 4 years.

NIH toolbox

Cognitive performance was assessed every second year (i.e., Y0, Y2, and Y4) using the cognition battery from the NIH Toolbox. This screen-based toolbox has been approved for preadolescents and includes a series of seven tasks that target different aspects of cognition (Weintraub, Bauer, et al., 2013). Each task has been validated for test-retest reliability and shows age-related effects (Luciana et al., 2018). A detailed review of the Toolbox and its tasks for the ABCD Study can be found in Luciana et al. (2018). Not all NIH Toolbox tasks were administered at each wave, and only tasks administered at all three waves were used in the current study. Uncorrected standard scores were used for cognitive abilities as recommended for longitudinal analyses by the ABCD Study (Neurocognition Data Release Notes, 2024).

Crystallized abilities: Reading and vocabulary

Crystallized ability was measured as an aggregate of uncorrected standard scores for oral reading and vocabulary knowledge performance. Reading performance was measured using the Oral Reading Recognition Task from the NIH Toolbox (Luciana et al., 2018). In this task, participants were asked to pronounce letters or words displayed on a screen. Words varied in difficulty, and uncommon words were included. Researchers listened to responses and scored each pronunciation as correct or incorrect (Gershon et al., 2014). Test-retest reliability was excellent for the three-wave average (ICC(3,3) consistency=0.88). Vocabulary knowledge was measured using the NIH Toolbox Picture Vocabulary Test (Luciana et al., 2018). This skill is known to be robustly associated with school performance (Gershon et al., 2014). For this task, participants heard audio recordings of words and were shown four images on a screen. From the four options, participants were instructed to select the image that best corresponded with the word (Luciana et al., 2018). Test-retest reliability was excellent (ICC(3,3) consistency=0.89). The two tasks were moderately correlated within each time point (correlation range from $r=.53$, $p<.001$ to $r=.59$, $p<.001$), supporting aggregation. Internal reliability of the two-indicator composite was good across waves ($\alpha_{Y0}=.69$, $\alpha_{Y2}=.73$, $\alpha_{Y4}=.73$). A z -score average of the two tasks anchored to Y0 was used for the main analyses to account for scaling differences.

Fluid abilities: Attention and processing speed

Fluid ability constructs available at all three time points included inhibitory control/attention, processing speed, and episodic memory. Previous studies have shown substantial ceiling effects for the episodic memory task, supporting its exclusion from analyses (Anokhin et al., 2022). Inhibitory control/attention was measured using the Eriksen Flanker Task (Eriksen & Eriksen, 1974). The NIH Toolbox Flanker Task measures the extent to which responses are impacted by ambient stimuli. For this task, youth were shown a series of blocks containing a row of five arrows, one middle arrow and four flanking arrows (Luciana et al., 2018). The orientation of the arrows varied, and the middle arrow was either facing the same direction (congruent) or a different direction from the flanking arrows (incongruent). For each block, the youth were instructed to push a button as fast as possible to indicate the direction of the middle arrow. Performance scores were based on speed and accuracy on both trial types (Luciana et al., 2018). Test-retest reliability was moderate (ICC(3,3) consistency=0.60). Processing speed was measured using the NIH Toolbox Pattern Comparison Processing Speed Test (Slotkin et al., 2012). In this task, participants were presented with pairs of images and asked to determine whether the two images were the same or different. The images were presented in pairs, and participants were given 90 s to respond to as many of the 130 pairs as possible. The final score was the number of correct comparisons discerned within the allotted 90 s. This task has been validated for adolescent samples (Carlozzi et al., 2013). Test-retest reliability was moderate (ICC(3,3) consistency=0.73). Correlations between fluid abilities were weak to moderate within time, from $r=.35$, $p<.001$ to $r=.45$, $p<.001$, and thus separate models were tested for attention/inhibition and processing speed.

Social media use

Youth reported on their social media use habits by answering questions about weekday and weekend hours (i.e., two questions in total) "On a typical [WEEKEND DAY/WEEKDAY], how many hours do you visit social networking sites like Facebook, Twitter, Instagram, etc.?" Responses were multiple choice and consisted of two items and were coded as follows: Y0 hours (1–4), Y2/Y4 (1–23), and minutes (0, 15, 30, or 45). Mean daily hours were calculated as a weighted average of weekend (2/7) and weekday (5/7) hours. Adolescents tend to overestimate their social media time in self-reports. However, at the between-person level, self-reported use shows acceptable convergent validity with device-logged totals ($\approx r=.55-.60$), meeting conventional thresholds for validity (Verbeij et al., 2021). Moreover, the predictive validity of self-reported use on developmental outcomes has been found to be comparable to digital trace measures in previous research (Verbeij et al., 2022). Accordingly, for our between-person aims (comparing adolescents with relatively higher vs. lower

use), self-reported time provides a defensible indicator of individual differences, while we avoid interpreting absolute hours as exact. Social media hours data were positively skewed. Therefore, this variable was log-transformed to reduce skewness and approximate a normal distribution.

Covariates

Youth sex and parental education level at baseline were included as time-invariant covariates in all models due to their known associations with screen habits and cognitive performance (Rindermann & Ceci, 2018; Totland et al., 2013; Twenge & Farley, 2021). Furthermore, the impact of other forms of electronic media use (i.e., video games, TV viewing, and online streaming) was taken into account by regressing slopes and intercepts on baseline levels of each media type.

Analytic plan

All analyses were conducted using Mplus version 8.3 (Muthén & Muthén, 2019). Missing data were handled with full-information maximum likelihood estimation using robust standard errors (MLRs) to account for the non-normality of social media hours (Yuan & Bentler, 2000). Descriptive statistics and bivariate correlations were calculated for all study variables.

To examine the concurrent longitudinal processes of cognitive abilities and social media use, three cognitive domain-specific parallel-process latent growth curve models were estimated. First, separate univariate latent growth curves were specified for each cognitive domain and social media use to assess model fit, estimating intercept and slope factors for each construct. The intercept factor represented individual differences at the initial assessment, whereas the slope factor captured rates of change across the three study waves. Slope loadings were rescaled to reflect assessment timing (i.e., Baseline, Year 2, and Year 4).

Second, growth in social media use was modeled concurrently with growth in each cognitive domain in three separate models. The parallel-process framework allows intercepts and slopes to covary, thereby testing associations between the initial levels and rates of change in cognitive ability and social media use. Intercept-intercept and slope-slope covariances were freely estimated, and slopes were regressed on intercepts to estimate effects while controlling for initial levels and covariates.

Third, after establishing adequate model fit, we included baseline video game hours, TV viewing hours, online streaming hours, youth sex, and parent education level as predictors of both intercepts and slopes to control for potential confounding effects. Model fit was evaluated using conventional benchmarks for fit indices (i.e., RMSEA < 0.06, SRMR < 0.08, CFI > 0.95, TLI > 0.95; Hu & Bentler, 1999). Adjusted *p*-values for cross-construct associations were

calculated to account for multiple comparisons using the Benjamini–Hochberg False Discovery Rate approach (Benjamini & Hochberg, 1995). The resulting models clarify the extent to which starting points and trajectories of cognitive abilities and social media hours are linked throughout the study period.

Missing data were minimal at baseline, moderate by Y2, and substantial by Y4, due to the ABCD Study's data release strategy (Y1: 1.3%–1.4%, Y3: 13.1%–32.0%, Y5: 61.1%–72.5%). Analyses of attrition patterns revealed that youth identifying as Black or Hispanic had higher odds of missingness at Y2 and Y4 (range OR = 1.30–1.69, $p < .001$), and higher parent education was related to lower odds of missingness at Y2 (range OR = 0.87–0.89, $p < .001$), but not at Y4. Because race and parental education were correlated with both social media hours and cognitive performances, these variables were included as covariates, and data were considered Missing At Random (MAR), which is compatible with MLR estimation (Enders & Bandalos, 2001). Finally, we conducted sensitivity analyses by re-estimating each model using listwise deletion to determine whether results were robust without using MLR estimation. The Mplus syntax and output for study analyses are available at <https://osf.io/5d9s8/>.

RESULTS

Descriptive statistics

Table 1 shows the means, standard deviations, and bivariate correlations of the study variables. As expected, the mean levels of social media hours and cognitive performance increased over the three waves. Cognitive within-domain correlations showed moderate to high positive correlations over time, supporting longitudinal analysis of these measures. The different cognitive skills were positively correlated at each time point with stronger associations between the Flanker and Pattern Comparison tasks than between fluid ability and crystallized performance, as expected. Further, social media hours tended to be weakly related to worse cognitive skills performance across domains within time points, associations that were particularly attenuated for fluid abilities.

Univariate growth models

The results of the four separate univariate growth curve are shown in Table 2. Each model showed excellent model fit (CFI ≥ 0.98, TLI ≥ 0.97, SRMR ≥ 0.02, RMSEA ≥ 0.05). For social media hours (Table 2, Model 1), the slope was significant and positive, suggesting that youth social media hours increased over time, on average. Moreover, social media hours intercept and slope were negatively associated ($i_{sm} \leftrightarrow s_{sm}$; $b = -0.20$, $p < .001$). For covariate effects on social media hours growth, being male ($\beta = -.18$, $p < .001$), higher parents' education ($\beta = -.12$, $p < .001$), and more Y0 video game hours

TABLE 1 Means, standard deviations, and bivariate correlations among study variables.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Parent education	–													
2. Sex (F = 0, M = 1)	.01	–												
3. TV EMU Y0	-.17***	.01	–											
4. VG EMU Y0	-.17***	.30***	.31***	–										
5. Stream EMU Y0	-.24***	.07***	.31***	.44***	–									
6. SMU Y0	-.13***	-.05***	.13***	.16***	.24***	–								
7. SMU Y2	-.22***	-.16***	.09***	.06***	.20***	.23***	–							
8. SMU Y4	-.22***	-.18***	.14***	.06***	.22***	.20***	.35***	–						
9. Crystallized Y0	.45***	.02	-.17***	-.12***	-.18***	-.13***	-.13***	-.19***	–					
10. Crystallized Y2	.45***	.02	-.18***	-.11***	-.18***	-.15***	-.16***	-.23***	.82***	–				
11. Crystallized Y4	.44***	.01	-.18***	-.11***	-.15***	-.13***	-.18***	-.24***	.78***	.82***	–			
12. Flanker Y0	.19***	.02	-.08***	-.05**	-.08***	-.03*	-.05*	-.04*	.31***	.30***	.26***	–		
13. Flanker Y2	.21***	.03*	-.12***	-.04*	-.08***	-.05**	-.08***	-.10***	.32***	.32***	.27***	.44***	–	
14. Flanker Y4	.24***	.08***	-.12***	-.02	-.06***	-.04*	-.06***	-.12***	.32***	.34***	.33***	.35***	.48***	–
15. Pattern Y0	.12***	-.06***	-.07***	-.06***	-.09***	-.04*	-.02	-.06***	.22***	.20***	.19***	.38***	.24***	.25***
16. Pattern Y2	.13***	-.08***	-.11***	-.07***	-.09***	-.04*	-.01	-.05**	.25***	.24***	.19***	.29***	.42***	.28***
17. Pattern Y4	.17***	-.08***	-.11***	-.06***	-.09***	.00	.01	-.04*	.26***	.26***	.23***	.29***	.32***	.47***
M	3.74	.52	1.26	1.01	.98	.12	.68	1.71	87.66	91.97	96.93	93.99	100.14	104.34
SD	1.17	.50	1.04	1.10	1.14	.42	1.32	2.05	6.59	6.81	9.15	9.14	7.71	7.31
Variable	15					16					17			
15. Pattern Y0	–					–					–			
16. Pattern Y2	.48***					.53***					–			
17. Pattern Y4	.41***					103.53					115.66			
M	88.06					15.09					16.73			
SD	14.59					15.09					16.73			

Note: M and SD are used to represent mean and standard deviation, respectively. * indicates $p < .05$. ** indicates $p < .01$. *** indicates $p < .001$.

Abbreviations: F, female; M, male; SMU, Social media use; Y, year.

TABLE 2 Univariate latent growth curve models parameter estimates.

Parameter	Est.	SE	p-Value	95% CI
Model 1: Social media use				
Intercept (i_{sm})	0.11	0.01	<.001	[0.10, 0.12]
Slope (s_{sm})	0.36	0.01	<.001	[0.35, 0.38]
Intercept variance	0.11	0.04	<.001	[0.03, 0.18]
Slope variance	0.13	0.02	<.001	[0.10, 0.17]
Intercept-slope covariance	-0.20	0.02	<.001	[-0.24, -0.15]
Model 2: Crystallized abilities				
Intercept (i_{cryst})	87.61	0.06	<.001	[87.49, 87.73]
Slope (s_{cryst})	2.13	0.01	<.001	[2.11, 2.16]
Intercept variance	36.59	0.77	<.001	[35.07, 38.11]
Slope variance	0.35	0.07	<.001	[0.18, 0.53]
Intercept-slope covariance	0.29	0.21	.170	[-0.13, 0.72]
Model 3: Flanker inhibitory control and attention task				
Intercept ($i_{flanker}$)	95.99	0.17	<.001	[95.66, 96.33]
Slope ($s_{flanker}$)	2.01	0.06	<.001	[1.89, 2.13]
Intercept variance	37.11	1.99	<.001	[33.21, 41.01]
Slope variance	1.04	0.26	<.001	[0.52, 1.55]
Intercept-slope covariance	-2.69	0.66	<.001	[-3.98, -1.41]
Model 4: Pattern matching processing speed task				
Intercept ($i_{pattern}$)	91.58	0.34	<.001	[90.91, 92.25]
Slope ($s_{pattern}$)	5.90	0.13	<.001	[5.64, 6.15]
Intercept variance	115.73	5.22	<.001	[105.50, 125.97]
Slope variance	5.52	0.87	<.001	[3.82, 7.21]
Intercept-slope covariance	-4.14	1.92	.032	[-7.91, -0.37]

Note: All models showed excellent fit (CFI $\geq$ 0.98, TLI $\geq$ 0.97, SRMR $\geq$ 0.02, RMSEA $\geq$ 0.05).

Abbreviations: Cryst, Crystallized task performance; SM, Social media.

($\beta = -.06$, $p = .001$) were related to lower social media hours slope, while being black was related to higher slope ($\beta = .15$, $p < .001$).

Crystallized abilities (Table 2, Model 2) showed significant positive growth across the study period and a nonsignificant intercept-slope covariance ($i_{cryst} \leftrightarrow s_{cryst}$: $b = 0.29$, $p = .170$), suggesting that youth increased these abilities on average, yet the magnitude of growth was independent of initial levels. Moreover, parents' education ($\beta = .16$, $p < .001$) and Y0 video games hours ($\beta = .06$, $p = .041$) were related to higher slope.

Flanker Inhibitory Control and Attention Task performance (Table 2, Model 3) demonstrated significant positive growth and a significant negative intercept-slope covariance ($i_{flanker} \leftrightarrow s_{flanker}$: $b = -2.69$, $p < .001$). This result indicates that youth with better Flanker performance at baseline tended to show less growth across the study period. Moreover, being male was related to a higher Flanker slope ($\beta = .08$, $p = .012$). However, parents' education and other forms of media use did not significantly predict Flanker performance.

Pattern Comparison Processing Speed Task performance (Table 2, Model 4) showed significant positive growth across time and a negative intercept-slope

association ($i_{pattern} \leftrightarrow s_{pattern}$: $b = -4.14$, $p = .032$). Moreover, parents' education ($\beta = .07$, $p = .017$) was positively related, and Y0 TV viewing hours ($\beta = -.07$, $p = .015$) was negatively related to the slope.

Crystallized performance parallel process model

Model fit for the crystallized performance vs. social media hours parallel process latent growth curve model was excellent (CFI = 0.99, TLI = 0.97, SRMR = 0.02, RMSEA = 0.03). Results for this model are presented in Table 3 (Model 5). A significant, albeit modest, negative intercept-intercept association between crystallized abilities and social media hours was found ($i_{cryst} \leftrightarrow i_{sm}$: $\beta = -.04$, $p = .024$, $p\text{-adj} = .041$). Consistent with hypotheses, initial levels of social media hours negatively predicted the slope of crystallized abilities ($i_{sm} \rightarrow s_{cryst}$: $\beta = -.22$, $p < .001$, $p\text{-adj} < .001$). This suggests that youths who initially used more social media demonstrated less growth in crystallized abilities over time. However, initial levels of crystallized ability were not related to the slope in social media hours ($i_{cryst} \rightarrow s_{sm}$: $\beta = .01$, $p = .665$, $p\text{-adj} = .665$). Lastly, as hypothesized, there

TABLE 3 Parallel process latent growth curve parameter estimates for each cognitive domain tested versus social media use.

Parameter	Est.	SE	β	p	p -adjusted	95% CI
Model 5: Crystallized abilities						
$i_{\text{cryst}} \leftrightarrow i_{\text{sm}}$	−0.19	0.02	−.04	.024	.041	[−0.33, 0.01]
$s_{\text{cryst}} \leftrightarrow s_{\text{sm}}$	−0.09	0.01	−.36	<.001	<.001	[−0.11, −0.07]
$i_{\text{cryst}} \rightarrow s_{\text{sm}}$	0.00	0.00	.01	.665	.665	[−0.01, 0.01]
$i_{\text{sm}} \rightarrow s_{\text{cryst}}$	−0.11	0.12	−.22	<.001	<.001	[−0.15, −0.07]
Model 6: Flanker inhibitory control and attention task						
$i_{\text{flanker}} \leftrightarrow i_{\text{sm}}$	0.39	0.12	.06	.001	.002	[0.16, 0.70]
$s_{\text{flanker}} \leftrightarrow s_{\text{sm}}$	−0.04	0.02	−.09	.031	.046	[−0.07, −0.01]
$i_{\text{flanker}} \rightarrow s_{\text{sm}}$	0.01	0.00	.04	.093	.124	[−0.01, 0.02]
$i_{\text{sm}} \rightarrow s_{\text{flanker}}$	−0.06	0.04	−.06	.115	.138	[−0.13, 0.04]
Model 7: Pattern matching processing speed task						
$i_{\text{pattern}} \leftrightarrow i_{\text{sm}}$	0.14	0.19	.01	.455	.496	[−0.23, 0.62]
$s_{\text{pattern}} \leftrightarrow s_{\text{sm}}$	0.12	0.04	.11	.001	.002	[0.05, 0.21]
$i_{\text{pattern}} \rightarrow s_{\text{sm}}$	0.02	0.00	.06	.001	.002	[0.01, 0.03]
$i_{\text{sm}} \rightarrow s_{\text{pattern}}$	0.22	0.08	.10	.003	.006	[0.08, 0.41]

Note: All models showed excellent fit ($\text{CFI} \geq 0.98$, $\text{TLI} \geq 0.97$, $\text{SRMR} \geq 0.03$, $\text{RMSEA} \geq 0.03$). Each model controlled for youth sex, race, parental education, Y0 tv hours, Y0 streaming hours, and Y0 video game hours. Covariates are not shown here but significant effects are reported in the text.

Abbreviations: CI, confidence interval; Cryst, crystallized task performance; SM, social media.

was a robust negative slope–slope association ($s_{\text{cryst}} \leftrightarrow s_{\text{sm}}$: $\beta = -.36$, $p < .001$) suggesting that an upward social media slope was related to decreased growth in crystallized abilities over time.

Flanker inhibitory control/attention performance parallel process model

Model fit for the Flanker task performance vs. social media hours model was excellent ($\text{CFI} = 0.99$, $\text{TLI} = 0.98$, $\text{SRMR} = 0.02$, $\text{RMSEA} = 0.03$). Results for this model are displayed in Table 3 (Model 6). The cross-construct association between intercepts of Flanker performance and social media hours was significant and positive ($i_{\text{flanker}} \leftrightarrow i_{\text{sm}}$: $\beta = .06$, $p = .001$, $p\text{-adj} = .002$). Further, initial levels of Flanker performance were not associated with the slope of social media hours ($i_{\text{flanker}} \rightarrow s_{\text{sm}}$: $\beta = .04$, $p = .093$, $p\text{-adj} = .124$). In addition, initial levels of social media hours were unrelated to the slope of fluid ability ($i_{\text{sm}} \rightarrow s_{\text{flanker}}$: $\beta = -.06$, $p = .115$, $p\text{-adj} = .138$). Last, the cross-construct association between slopes was significant and negative ($s_{\text{flanker}} \leftrightarrow s_{\text{sm}}$: $\beta = -.09$, $p = .031$, $p\text{-adj} = .046$), indicating potential trade-offs.

Pattern comparison processing speed performance parallel process model

Model 7 in Table 2 shows growth in Pattern Comparison task performance vs. social media hours associations. Model fit was excellent ($\text{CFI} = 0.99$, $\text{TLI} = 0.98$, $\text{SRMR} = 0.01$, $\text{RMSEA} = 0.04$). Although the intercepts were unrelated ($i_{\text{pattern}} \leftrightarrow i_{\text{sm}}$: $\beta = .01$, $p = .455$, $p\text{-adj} = .496$), both intercepts

predicted positive cross-construct growth ($i_{\text{sm}} \rightarrow s_{\text{pattern}}$: $\beta = .10$, $p = .003$, $p\text{-adj} = .006$; $i_{\text{pattern}} \rightarrow s_{\text{sm}}$: $\beta = .06$, $p = .001$, $p\text{-adj} = .002$). Moreover, slope–slope associations were positively related ($s_{\text{pattern}} \leftrightarrow s_{\text{sm}}$: $\beta = .11$, $p = .001$, $p\text{-adj} = .002$). Taken together, these results suggest that processing speed and social media hours are positively coupled throughout early adolescence.

Sensitivity analyses

All models were also tested using listwise deletion ($N = 4289$) to test the robustness of results without MLR estimation due to high levels of missing data in the final time point. Nearly all key findings, including all slope–slope associations, remained similar in magnitude, significance level, and direction. However, for crystallized abilities, the intercept–intercept association with social media hours was no longer significant. Moreover, the cross-construct intercept–slope associations were no longer significant in the processing speed versus social media hours model. Overall, these sensitivity analyses suggest that trajectory associations were robust to listwise deletion; however, some associations with smaller effect sizes involving the intercepts were under powered due to the reduced sample size.

DISCUSSION

The present study examined longitudinal associations between social media engagement and cognitive development across early adolescence. Using parallel-process latent growth curve models, we tested whether trajectories of social

media use covaried with trajectories of cognitive abilities spanning crystallized and fluid domains. Results revealed domain-specific coupling patterns. First, greater increases in social media hours were associated with slower growth in crystallized abilities. However, associations for fluid abilities were task-specific. Social media and inhibitory control/attention trajectories were inversely related, whereas social media and processing speed trajectories were positively linked. Together, these findings suggest that social media engagement relates to distinct aspects of cognitive development in nuanced, task-specific ways.

Consistent with Hypothesis 1, growth in social media hours was negatively associated with growth in crystallized abilities, indicating that adolescents who increased their social media engagement more rapidly also showed slower gains in vocabulary and acquired knowledge over time. It may be that increasing exposure to rapidly shifting, attention-fragmented online environments (Junco, 2012a) reinforces superficial processing styles that interfere with sustained focus (Stothart et al., 2015; Ward et al., 2017), impacting verbal knowledge acquisition (Wei et al., 2012). This pattern also aligns with prior evidence for time-displacement processes (Twenge et al., 2019). In particular, social media use may reduce opportunities for cognitively enriching activities such as reading, structured study, and parent-child conversation that are known to support crystallized growth (Garcia-Contiente et al., 2013; Zahar et al., 2001). In the reverse direction, greater crystallized growth and academic success may reinforce engagement in these enriching activities, reducing opportunities for social media engagement and amplifying developmental divergence over time. Explicit directional models are needed to disentangle directionality in these findings.

Test results further showed that, although higher initial social media use predicted slower subsequent crystallized growth, the reverse effect was not observed, suggesting that early social media adoption was associated with longer-term crystallized ability development. This finding builds on prior literature linking early social media initiation with unhealthy and compulsive online behavior in later adolescence (Charmaraman et al., 2022). Because crystallized skills build cumulatively throughout adolescence (Thorsen et al., 2014), even modest early slowdowns in growth may have lasting academic and functional implications.

In contrast to Hypothesis 2a, our results showed a weak negative association between social media growth and inhibitory control/attention growth. This finding extends prior cross-sectional studies that showed modest negative correlations between fluid abilities and media use in adolescence (Cain et al., 2016; Lara & Bokoch, 2021; Stieger & Wunderl, 2022) by highlighting that these constructs co-develop inversely across early adolescence. The pattern may reflect a trade-off between exposure to fast-paced, notification-driven digital environments and the maturation of sustained attention. Conversely, it may be that youth with weaker attentional control may gravitate toward social media because it provides immediate stimulation with minimal sustained effort, consistent with evidence

linking impulsivity to greater screen engagement (Grund & Luciana, 2025).

Intercept-slope associations between social media hours and inhibitory control/attention showed no evidence of baseline effects, suggesting that early differences in attentional control did not determine subsequent social media use or vice versa. Instead, the association appears to emerge developmentally, as both constructs were negatively linked at baseline and their slopes remained so across early adolescence. This pattern implies that social media engagement and inhibitory control may become coupled through dynamic interacting processes, such as growing autonomy, peer influences, and individual traits, that shape both attentional development and digital behavior. Alternatively, individual differences in unmeasured environmental factors could contribute to parallel changes in both trajectories.

In contrast with Hypothesis 2b, trajectories of social media use and processing speed were positively associated, suggesting a potential domain-specific benefit of social media use. This result may reflect practice-based transfer (Bavelier et al., 2012; Dye et al., 2009), as frequent digital interactions (e.g., rapid visual scanning, stimulus-response mapping, and fine-motor execution while typing or scrolling) mirror the perceptual-motor demands of the Pattern Comparison task. Alternatively, youth with faster baseline processing speed may find social media more engaging or easier to navigate, reinforcing continued use, a selection mechanism rather than causal enhancement (Uncapher & Wagner, 2018). The presence of positive cross-domain baseline effects in both directions further supports potential bidirectional associations between processing speed and social media hours. Because these effects were specific to processing speed and not observed for inhibitory control, they likely represent task-specific overlap in underlying micro-skills rather than broad cognitive improvement due to social media exposure.

Collectively, these domain-specific patterns indicate that social media use is not uniformly detrimental or beneficial for cognition but interacts with cognitive development in differentiated ways. Crystallized skills, which depend on cumulative verbal and conceptual learning, appear most vulnerable to social media effects. In contrast, aspects of fluid performance tied to rapid perceptual processing may co-develop with social media engagement, whereas inhibitory control/attention exhibits potential interference.

Limitations and future directions

Several limitations qualify these conclusions. First, the social media measure did not distinguish between passive use (e.g., scrolling) and active use (e.g., posting), preventing differentiation between potentially beneficial and detrimental engagement. More nuanced assessments could clarify whether content, frequency, or interaction type drives observed associations. Second, although the models reveal longitudinal coupling, causal inferences remain tentative

without experimental or quasi-experimental designs. Third, social media hours were self-reported; integrating objective indicators such as passive sensing apps would improve validity, though prior work suggests youth self-reports are more accurate than parent reports (Wade et al., 2021). Fourth, the crystallized intelligence measure was limited to reading and vocabulary, and future studies should examine additional knowledge domains (e.g., quantitative or cultural). Fifth, although parental education was accounted for, unmeasured confounders such as baseline IQ could influence both social media use and cognitive performance. Including IQ as a covariate in future work would strengthen inference. Sixth, while trajectories of social media hours and processing speed were positively linked, it remains unclear whether this association extends beyond screen-based contexts. Finally, although the NIH Toolbox is widely used for youth cognition (Weintraub, Bauer, et al., 2013; Weintraub, Dikmen, et al., 2013), concerns about its temporal reliability within large multisite designs (Taylor et al., 2022) suggest that low measurement stability may have attenuated observed effects (Fitzmaurice et al., 2012, p. 40; Schmidt & Hunter, 1999). Replication using more reliable instruments is warranted.

CONCLUSION

Despite these design limitations, this study leveraged a large, representative sample to longitudinally examine the associations between social media hours and task-based measures of cognitive functioning. Our results highlight that while youth generally gain in both knowledge-based and executive capacities, social media growth was robustly coupled with hindered improvement in crystallized abilities. This is perhaps because crystallized skills draw directly from environmental exposure and can be impacted by reading habits and other activities. In contrast, social media and fluid ability coupling was domain-specific, with social media trajectories positively coupled with processing speed, but negatively coupled with inhibitory control/attention. These findings have implications for a broad swath of the population as social media is one of the most common leisure activities for adolescent youth. Moreover, because disadvantaged youth tend to use more screens than wealthier peers (Tandon et al., 2012), they may experience an outsized impact, contributing to a further widening academic achievement gap based on socioeconomic status. For parents and professionals, emphasizing exposure to more advanced language content and encouraging moderation in social media behavior in the youth under their care may be beneficial for maintaining desired trajectories in crystallized and inhibition/attentional capacities growth.

AUTHOR CONTRIBUTIONS

Niyantri Ravindran: Conceptualization; writing – review and editing; methodology; supervision. **Cory Carvalho:** Conceptualization; writing – review and editing; methodology; data curation; analysis.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available to qualified researchers with a valid data use contract in the NIMH Data Archive (NDA) at <https://nda.nih.gov>, NDA Collection #2573, DOI: [10.15154/z563-zd24](https://doi.org/10.15154/z563-zd24).

ETHICS STATEMENT

This study involved secondary analyses of existing, de-identified data from the Adolescent Brain Cognitive Development (ABCD) Study, which were obtained from the National Institute of Mental Health Data Archive (NDA). All procedures for data collection in the ABCD Study were reviewed and approved by the Institutional Review Board (IRB) at the University of California, San Diego (UC San Diego), and by the IRBs of each individual study site. All methods were carried out in accordance with relevant guidelines and regulations, and the research team adhered to the ABCD Study data use agreement, ensuring participant confidentiality and data security.

PARTICIPANT CONSENT STATEMENT

Informed consent and assent were obtained from all ABCD Study participants and/or their legal guardians at the time of

enrollment. All consenting participants have further given consent for data sharing for secondary data analysis.

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REFERENCES

- Akshoomoff, N., Beaumont, J. L., Bauer, P. J., Dikmen, S. S., Gershon, R. C., Mungas, D., Slotkin, J., Tulskey, D., Weintraub, S., Zelazo, P. D., & Heaton, R. K. (2013). VIII. NIH toolbox cognition battery (CB): Composite scores of crystallized, fluid, and overall cognition. *Monographs of the Society for Research in Child Development*, 78(4), 119–132. <https://doi.org/10.1111/mono.12038>
- Alloway, T. P., Horton, J., Alloway, R. G., & Dawson, C. (2013). Social network working sites and cognitive abilities: Do they make you smarter? *Computers & Education*, 63, 10–16. <https://doi.org/10.1016/j.compedu.2012.10.030>
- Almarzouki, A. F., Alghamdi, R. A., Nassar, R., Aljohani, R. R., Nasser, A., Bawadood, M., & Almalki, R. H. (2022). Social media usage, working memory, and depression: An experimental investigation among university students. *Behavioral Science*, 12(1), 16. <https://doi.org/10.3390/bs12010016>
- Anokhin, A. P., Luciana, M., Banich, M., Barch, D., Bjork, J. M., Gonzalez, M. R., Gonzalez, R., Haist, F., Jacobus, J., Lisdahl, K., McGlade, E., McCandliss, B., Nagel, B., Nixon, S. J., Tapert, S., Kennedy, J. T., & Thompson, W. (2022). Age-related changes and longitudinal stability of individual differences in ABCD neurocognition measures. *Developmental Cognitive Neuroscience*, 54, 101078. <https://doi.org/10.1016/j.dcn.2022.101078>
- Bavelier, D., Green, C. S., Pouget, A., & Schrater, P. (2012). Brain plasticity through the life span: Learning to learn and action video games. *Annual Review of Neuroscience*, 35, 391–416. <https://doi.org/10.1146/annurev-neuro-060909-152832>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B, Statistical Methodology*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Blair, C. (2006). How similar are fluid cognition and general intelligence? A developmental neuroscience perspective on fluid cognition as an aspect of human cognitive ability. *Behavioral and Brain Sciences*, 29(2), 109–125. <https://doi.org/10.1017/S0140525X06009034>
- Blakemore, S.-J., & Choudhury, S. (2006). Development of the adolescent brain: Implications for executive function and social cognition. *Journal of Child Psychology and Psychiatry*, 47(3–4), 296–312. <https://doi.org/10.1111/j.1469-7610.2006.01611.x>
- Brydges, C. R., Ozolnieks, K. L., & Roberts, G. (2017). Working memory – Not processing speed – Mediates fluid intelligence deficits associated with attention deficit/hyperactivity disorder symptoms. *Journal of Neuropsychology*, 11(3), 362–377. <https://doi.org/10.1111/jnp.12096>
- Brydges, C. R., Ozolnieks, K. L., & Roberts, G. (2021). Working memory and intraindividual variability in response time mediate fluid intelligence deficits associated with ADHD symptomatology. *Journal of Attention Disorders*, 25(1), 63–72. <https://doi.org/10.1177/1087054718772143>
- Cain, M. S., Leonard, J. A., Gabrieli, J. D. E., & Finn, A. S. (2016). Media multitasking in adolescence. *Psychonomic Bulletin & Review*, 23(6), 1932–1941. <https://doi.org/10.3758/s13423-016-1036-3>
- Carlozzi, N. E., Tulskey, D. S., Kail, R. V., & Beaumont, J. L. (2013). VI. NIH toolbox cognition battery (CB): Measuring processing speed. *Monographs of the Society for Research in Child Development*, 78(4), 88–102. <https://doi.org/10.1111/mono.12036>
- Casey, B. J., Cannonier, T., Conley, M. I., Cohen, A. O., Barch, D. M., Heitzeg, M. M., Soules, M. E., Teslovich, T., Dellarco, D. V., Garavan, H., Orr, C. A., Wager, T. D., Banich, M. T., Speer, N. K., Sutherland, M. T., Riedel, M. C., Dick, A. S., Bjork, J. M., Thomas, K. M., ... Dale, A. M. (2018). The adolescent brain cognitive development (ABCD) study: Imaging acquisition across 21 sites. *Developmental Cognitive Neuroscience*, 32, 43–54. <https://doi.org/10.1016/j.dcn.2018.03.001>
- Cattell, R. B. (1987). *Intelligence: Its structure, growth and action*. Elsevier.
- Charmaraman, L., Lynch, A. D., Richer, A. M., & Grossman, J. M. (2022). Associations of early social media initiation on digital behaviors and the moderating role of limiting use. *Computers in Human Behavior*, 127, 107053. <https://doi.org/10.1016/j.chb.2021.107053>
- Cingel, D. P., & Sundar, S. S. (2012). Texting, techspeak, and tweens: The relationship between text messaging and English grammar skills. *New Media & Society*, 14(8), 1304–1320. <https://doi.org/10.1177/1461448124442927>
- Dye, M. W. G., Green, C. S., & Bavelier, D. (2009). Increasing speed of processing with action video games. *Current Directions in Psychological Science*, 18(6), 321–326. <https://doi.org/10.1111/j.1467-8721.2009.01660.x>
- Enders, C. K., & Bandalos, D. L. (2001). The relative performance of full information maximum likelihood estimation for missing data in structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, 8(3), 430–457. https://doi.org/10.1207/S15328007SEM0803_5
- Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. *Perception & Psychophysics*, 16(1), 143–149. <https://doi.org/10.3758/BF03203267>
- Ferschmann, L., Bos, M. G. N., Herting, M. M., Mills, K. L., & Tammes, C. K. (2022). Contextualizing adolescent structural brain development: Environmental determinants and mental health outcomes. *Current Opinion in Psychology*, 44, 170–176. <https://doi.org/10.1016/j.copsyc.2021.09.014>
- Fitzmaurice, G. M., Laird, N. M., & Ware, J. H. (2012). *Applied longitudinal analysis*. John Wiley & Sons.
- Freeman, J. (1983). Environment and high IQ—A consideration of fluid and crystallized intelligence. *Personality and Individual Differences*, 4(3), 307–313. [https://doi.org/10.1016/0191-8869\(83\)90152-6](https://doi.org/10.1016/0191-8869(83)90152-6)
- Garcia-Contiente, X., Pérez-Giménez, A., Espelt, A., & Nebot Adell, M. (2013). Factors associated with media use among adolescents: A multilevel approach. *European Journal of Public Health*, 24(1), 5–10. <https://doi.org/10.1093/eurpub/ckt013>
- Gershon, R. C., Cook, K. F., Mungas, D., Manly, J. J., Slotkin, J., Beaumont, J. L., & Weintraub, S. (2014). Language measures of the NIH toolbox cognition battery. *Journal of the International Neuropsychological Society*, 20(6), 642–651. <https://doi.org/10.1017/S1355617714000411>
- Gou, S., Yuan, R., Zhang, W., Tang, Y., & Zhang, W. (2024). Problematic social media use in youths cause response inhibition impairment. *Current Psychology*, 43(15), 13555–13564. <https://doi.org/10.1007/s12144-023-05425-z>
- Grund, B. A., & Luciana, M. (2025). Prospective predictors of adolescent screen time and problematic screen use. *JAACAP Open*, 3, 1259–1269. <https://doi.org/10.1016/j.jaacop.2025.06.007>
- Heeringa, S. G., & Berglund, P. A. (2020). A guide for population-based analysis of the adolescent brain cognitive development (ABCD) study baseline data. *bioRxiv*, 2020.2002.2010.942011. <https://doi.org/10.1101/2020.02.10.942011>
- Hoekstra, R. A., Bartels, M., & Boomsma, D. I. (2007). Longitudinal genetic study of verbal and nonverbal IQ from early childhood to young adulthood. *Learning and Individual Differences*, 17(2), 97–114. <https://doi.org/10.1016/j.lindif.2007.05.005>
- Horn, J. L. (1968). Organization of abilities and the development of intelligence. *Psychological Review*, 75(3), 242–259. <https://doi.org/10.1037/h0025662>
- Horn, J. L., & Cattell, R. B. (1966). Refinement and test of the theory of fluid and crystallized general intelligences. *Journal of Educational Psychology*, 57(5), 253–270. <https://doi.org/10.1037/h0023816>
- Horvitz, E., & Cutrell, E. (2001). Notification, disruption, and memory: Effects of messaging interruptions on memory and performance. *Human-Computer Interaction*.
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives.

- Structural Equation Modeling: A Multidisciplinary Journal*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- Junco, R. (2012a). In-class multitasking and academic performance. *Computers in Human Behavior*, 28(6), 2236–2243. <https://doi.org/10.1016/j.chb.2012.06.031>
- Junco, R. (2012b). Too much face and not enough books: The relationship between multiple indices of Facebook use and academic performance. *Computers in Human Behavior*, 28(1), 187–198. <https://doi.org/10.1016/j.chb.2011.08.026>
- Kan, K.-J., Wicherts, J. M., Dolan, C. V., & van der Maas, H. L. J. (2013). On the nature and nurture of intelligence and specific cognitive abilities: The more heritable, the more culture dependent. *Psychological Science*, 24(12), 2420–2428. <https://doi.org/10.1177/0956797613493292>
- Kennedy, J., & Lynch, H. (2016). A shift from offline to online: Adolescence, the internet and social participation. *Journal of Occupational Science*, 23(2), 156–167. <https://doi.org/10.1080/14427591.2015.1117523>
- Kim, S. A. (2017). Social media algorithms: Why you see what you see. *Georgetown Law Technology Review*, 2, 147–153.
- Kirschner, P. A., & Karpinski, A. C. (2010). Facebook® and academic performance. *Computers in Human Behavior*, 26(6), 1237–1245. <https://doi.org/10.1016/j.chb.2010.03.024>
- Lara, R. S., & Bokoch, R. (2021). Cognitive functioning and social media: Has technology changed us? *Acta Psychologica*, 221, 103429. <https://doi.org/10.1016/j.actpsy.2021.103429>
- Larsen, B., & Luna, B. (2018). Adolescence as a neurobiological critical period for the development of higher-order cognition. *Neuroscience & Biobehavioral Reviews*, 94, 179–195. <https://doi.org/10.1016/j.neubiorev.2018.09.005>
- Liu, J., Riesch, S., Tien, J., Lipman, T., Pinto-Martin, J., & O'Sullivan, A. (2022). Screen media overuse and associated physical, cognitive, and emotional/behavioral outcomes in children and adolescents: An integrative review. *Journal of Pediatric Health Care*, 36(2), 99–109. <https://doi.org/10.1016/j.pedhc.2021.06.003>
- Loughnan, R. J., Palmer, C. E., Thompson, W. K., Dale, A. M., Jernigan, T. L., & Chieh Fan, C. (2023). Intelligence polygenic score is more predictive of crystallized measures: Evidence from the adolescent brain cognitive development (ABCD) study. *Psychological Science*, 34(6), 714–725. <https://doi.org/10.1177/09567976231160702>
- Luciana, M. (2013). Adolescent brain development in normality and psychopathology. *Development and Psychopathology*, 25(4pt2), 1325–1345. <https://doi.org/10.1017/S0954579413000643>
- Luciana, M., Bjork, J. M., Nagel, B. J., Barch, D. M., Gonzalez, R., Nixon, S. J., & Banich, M. T. (2018). Adolescent neurocognitive development and impacts of substance use: Overview of the adolescent brain cognitive development (ABCD) baseline neurocognition battery. *Developmental Cognitive Neuroscience*, 32, 67–79. <https://doi.org/10.1016/j.dcn.2018.02.006>
- Madigan, S., Browne, D., Racine, N., Mori, C., & Tough, S. (2019). Association between screen time and children's performance on a developmental screening test. *JAMA Pediatrics*, 173(3), 244–250. <https://doi.org/10.1001/jamapediatrics.2018.5056>
- McGrew, K. S. (2009). CHC theory and the human cognitive abilities project: Standing on the shoulders of the giants of psychometric intelligence research. *Intelligence*, 37(1), 1–10. <https://doi.org/10.1016/j.intell.2008.08.004>
- Murty, V. P., Calabro, F., & Luna, B. (2016). The role of experience in adolescent cognitive development: Integration of executive, memory, and mesolimbic systems. *Neuroscience & Biobehavioral Reviews*, 70, 46–58. <https://doi.org/10.1016/j.neubiorev.2016.07.034>
- Muthén, B., & Muthén, L. (2019). *Mplus: A general latent variable modeling program*. Muthén & Muthén.
- Neophytou, E., Manwell, L. A., & Eikelboom, R. (2021). Effects of excessive screen time on neurodevelopment, learning, memory, mental health, and neurodegeneration: A scoping review. *International Journal of Mental Health and Addiction*, 19(3), 724–744. <https://doi.org/10.1007/s11469-019-00182-2>
- Neurocognition Data Release Notes. (2024). <https://wiki.abcdstudy.org/release-notes/non-imaging/neurocognition.html>
- Nisbett, R. E., Aronson, J., Blair, C., Dickens, W., Flynn, J., Halpern, D. F., & Turkheimer, E. (2012). Intelligence: New findings and theoretical developments. *American Psychologist*, 67(2), 130–159. <https://doi.org/10.1037/a0026699>
- Paus, T. (2005). Mapping brain maturation and cognitive development during adolescence. *Trends in Cognitive Sciences*, 9(2), 60–68. <https://doi.org/10.1016/j.tics.2004.12.008>
- Plomin, R., Haworth, C. M. A., Meaburn, E. L., Price, T. S., & Davis, O. S. P. (2013). Common DNA markers can account for more than half of the genetic influence on cognitive abilities. *Psychological Science*, 24(4), 562–568. <https://doi.org/10.1177/0956797612457952>
- Procopio, F., Zhou, Q., Wang, Z., Gidziela, A., Rimfeld, K., Malanchini, M., & Plomin, R. (2022). The genetics of specific cognitive abilities. *Intelligence*, 95, 101689. <https://doi.org/10.1016/j.intell.2022.101689>
- Reed, P. (2023). Impact of social media use on executive function. *Computers in Human Behavior*, 141, 107598. <https://doi.org/10.1016/j.chb.2022.107598>
- Rideout, V., Peebles, A., Mann, S., & Robb, M. B. (2022). *Common Sense census: Media use by tweens and teens, 2021*. C. Sense.
- Rindermann, H., & Ceci, S. J. (2018). Parents' education is more important than their wealth in shaping their children's intelligence: Results of 19 samples in seven countries at different developmental levels. *Journal for the Education of the Gifted*, 41(4), 298–326. <https://doi.org/10.1177/0162353218799481>
- Sauce, B., Liebherr, M., Judd, N., & Klingberg, T. (2022). The impact of digital media on children's intelligence while controlling for genetic differences in cognition and socioeconomic background. *Scientific Reports*, 12(1), 7720. <https://doi.org/10.1038/s41598-022-11341-2>
- Schmidt, F. L., & Hunter, J. E. (1999). Theory testing and measurement error. *Intelligence*, 27(3), 183–198. [https://doi.org/10.1016/S0160-2896\(99\)00024-0](https://doi.org/10.1016/S0160-2896(99)00024-0)
- Schneider, W. J., & McGrew, K. S. (2012). The Cattell-Horn-Carroll model of intelligence. In D. P. Flanagan & E. M. McDonough (Eds.), *Contemporary intellectual assessment: Theories, tests, and issues* (3rd ed., pp. 99–144). The Guilford Press.
- Schroeders, U., Schipolowski, S., & Wilhelm, O. (2015). Age-related changes in the mean and covariance structure of fluid and crystallized intelligence in childhood and adolescence. *Intelligence*, 48, 15–29. <https://doi.org/10.1016/j.intell.2014.10.006>
- Slotkin, J., Kallen, M., Griffith, J., Magasi, S., Salsman, J., Nowinski, C., & Gershon, R. C. (2012). *NIH toolbox technical manual*. https://www.healthmeasures.net/images/nihtoolbox/Technical_Manuals/Cognition/Toolbox_Pattern_Comparison_Processing_Speed_Test_Technical_Manual.pdf
- Spear, L. P. (2013). Adolescent neurodevelopment. *Journal of Adolescent Health*, 52(2, Supplement 2), S7–S13. <https://doi.org/10.1016/j.jadohealth.2012.05.006>
- Steinberg, L. (2005). Cognitive and affective development in adolescence. *Trends in Cognitive Sciences*, 9(2), 69–74. <https://doi.org/10.1016/j.tics.2004.12.005>
- Stieger, S., & Wunderl, S. (2022). Associations between social media use and cognitive abilities: Results from a large-scale study of adolescents. *Computers in Human Behavior*, 135, 107358. <https://doi.org/10.1016/j.chb.2022.107358>
- Stothart, C., Mitchum, A., & Yehnert, C. (2015). The attentional cost of receiving a cell phone notification. *Journal of Experimental Psychology: Human Perception and Performance*, 41(4), 893–897. <https://doi.org/10.1037/xhp0000100>
- Tandon, P. S., Zhou, C., Sallis, J. F., Cain, K. L., Frank, L. D., & Saelens, B. E. (2012). Home environment relationships with children's physical activity, sedentary time, and screen time by socioeconomic status. *International Journal of Behavioral Nutrition and Physical Activity*, 9(1), 88. <https://doi.org/10.1186/1479-5868-9-88>
- Taylor, B. K., Frenzel, M. R., Eastman, J. A., Wiesman, A. I., Wang, Y.-P., Calhoun, V. D., Stephen, J. M., & Wilson, T. W. (2022). Reliability of the NIH toolbox cognitive battery in children and adolescents: A 3-year longitudinal examination. *Psychological Medicine*, 52(9), 1718–1727. <https://doi.org/10.1017/S0033291720003487>

- Thorell, L. B., Burén, J., Ström Wiman, J., Sandberg, D., & Nutley, S. B. (2022). Longitudinal associations between digital media use and ADHD symptoms in children and adolescents: A systematic literature review. *European Child & Adolescent Psychiatry*, 33, 2503–2526. <https://doi.org/10.1007/s00787-022-02130-3>
- Thorsen, C., Gustafsson, J.-E., & Cliffordson, C. (2014). The influence of fluid and crystallized intelligence on the development of knowledge and skills. *British Journal of Educational Psychology*, 84(4), 556–570. <https://doi.org/10.1111/bjep.12041>
- Totland, T. H., Bjelland, M., Lien, N., Bergh, I. H., Gebremariam, M. K., Grydeland, M., Ommundsen, Y., & Andersen, L. F. (2013). Adolescents' prospective screen time by gender and parental education, the mediation of parental influences. *International Journal of Behavioral Nutrition and Physical Activity*, 10(1), 89. <https://doi.org/10.1186/1479-5868-10-89>
- Tsitsika, A. K., Tzavela, E. C., Janikian, M., Ólafsson, K., Iordache, A., Schoenmakers, T. M., Tzavara, C., & Richardson, C. (2014). Online social networking in adolescence: Patterns of use in six European countries and links with psychosocial functioning. *Journal of Adolescent Health*, 55(1), 141–147. <https://doi.org/10.1016/j.jadohealth.2013.11.010>
- Tucker-Drob, E. M., & Briley, D. A. (2014). Continuity of genetic and environmental influences on cognition across the life span: A meta-analysis of longitudinal twin and adoption studies. *Psychological Bulletin*, 140(4), 949–979. <https://doi.org/10.1037/a0035893>
- Twenge, J. M., & Farley, E. (2021). Not all screen time is created equal: Associations with mental health vary by activity and gender. *Social Psychiatry and Psychiatric Epidemiology*, 56(2), 207–217. <https://doi.org/10.1007/s00127-020-01906-9>
- Twenge, J. M., Martin, G. N., & Spitzberg, B. H. (2019). Trends in US adolescents' media use, 1976–2016: The rise of digital media, the decline of TV, and the (near) demise of print. *Psychology of Popular Media Culture*, 8(4), 329.
- Uncapher, M. R., & Wagner, A. D. (2018). Minds and brains of media multitaskers: Current findings and future directions. *Proceedings of the National Academy of Sciences of the United States of America*, 115(40), 9889–9896. <https://doi.org/10.1073/pnas.1611612115>
- van Soelen, I. L. C., Brouwer, R. M., Leeuwen, M. v., Kahn, R. S., Hulshoff Pol, H. E., & Boomsma, D. I. (2011). Heritability of verbal and performance intelligence in a pediatric longitudinal sample. *Twin Research and Human Genetics*, 14(2), 119–128. <https://doi.org/10.1375/twin.14.2.119>
- Verbeij, T., Pouwels, J. L., Beyens, I., & Valkenburg, P. M. (2021). The accuracy and validity of self-reported social media use measures among adolescents. *Computers in Human Behavior Reports*, 3, 100090. <https://doi.org/10.1016/j.chbr.2021.100090>
- Verbeij, T., Pouwels, J. L., Beyens, I., & Valkenburg, P. M. (2022). Experience sampling self-reports of social media use have comparable predictive validity to digital trace measures. *Scientific Reports*, 12(1), 7611. <https://doi.org/10.1038/s41598-022-11510-3>
- Wade, N. E., Ortigara, J. M., Sullivan, R. M., Tomko, R. L., Breslin, F. J., Baker, F. C., Fuemmeler, B. F., Delrahim Howlett, K., Lisdahl, K. M., Marshall, A. T., Mason, M. J., Neale, M. C., Squeglia, L. M., Wolff-Hughes, D. L., Tapert, S. F., & Bagot, K. S. (2021). Passive sensing of preteens' smartphone use: An adolescent brain cognitive development (ABCD) cohort substudy. *JMIR Mental Health*, 8(10), e29426. <https://doi.org/10.2196/29426>
- Walsh, J. J., Barnes, J. D., Tremblay, M. S., & Chaput, J.-P. (2020). Associations between duration and type of electronic screen use and cognition in US children. *Computers in Human Behavior*, 108, 106312. <https://doi.org/10.1016/j.chb.2020.106312>
- Ward, A. F., Duke, K., Gneezy, A., & Bos, M. W. (2017). Brain drain: The mere presence of one's own smartphone reduces available cognitive capacity. *Journal of the Association for Consumer Research*, 2(2), 140–154. <https://doi.org/10.1086/691462>
- Wei, F.-Y. F., Wang, Y. K., & Klausner, M. (2012). Rethinking college students' self-regulation and sustained attention: Does text messaging during class influence cognitive learning? *Communication Education*, 61(3), 185–204. <https://doi.org/10.1080/03634523.2012.672755>
- Weintraub, S., Bauer, P. J., Zelazo, P. D., Wallner-Allen, K., Dikmen, S. S., Heaton, R. K., Tulsky, D. S., Slotkin, J., Blitz, D. L., Carlozzi, N. E., Havlik, R. J., Beaumont, J. L., Mungas, D., Manly, J. J., Borosh, B. G., Nowinski, C. J., & Gershon, R. C. (2013). I. NIH toolbox cognition battery (CB): Introduction and pediatric data. *Monographs of the Society for Research in Child Development*, 78(4), 1–15. <https://doi.org/10.1111/mono.12031>
- Weintraub, S., Dikmen, S. S., Heaton, R. K., Tulsky, D. S., Zelazo, P. D., Bauer, P. J., Carlozzi, N. E., Slotkin, J., Blitz, D., Wallner-Allen, K., Fox, N. A., Beaumont, J. L., Mungas, D., Nowinski, C. J., Richler, J., Deocampo, J. A., Anderson, J. E., Manly, J. J., Borosh, B., ... Gershon, R. C. (2013). Cognition assessment using the NIH toolbox. *Neurology*, 80(11 s3), S54–S64. <https://doi.org/10.1212/WNL.0b013e3182872ded>
- Wilmer, H. H., Sherman, L. E., & Chein, J. M. (2017). Smartphones and cognition: A review of research exploring the links between mobile technology habits and cognitive functioning [Review]. *Frontiers in Psychology*, 8, 605. <https://doi.org/10.3389/fpsyg.2017.00605>
- Wood, S. J., De Luca, C. R., Anderson, V., & Pantelis, C. (2004). Cognitive development in adolescence: Cerebral underpinnings, neural trajectories and the impact of aberrations. In M. S. Keshavan, J. L. Kennedy, & R. M. Murray (Eds.), *Neurodevelopment and schizophrenia* (Vol. 1, pp. 69–88). Cambridge University Press.
- Yuan, K.-H., & Bentler, P. M. (2000). Three likelihood-based methods for mean and covariance structure analysis with nonnormal missing data. *Sociological Methodology*, 30(1), 165–200. <https://doi.org/10.1111/0081-1750.00078>
- Zahar, R., Cobb, T., & Spada, N. (2001). Acquiring vocabulary through reading: Effects of frequency and contextual richness. *The Canadian Modern Language Review*, 57(4), 541–572. <https://doi.org/10.3138/cmlr.57.4.541>

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